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Turbo Machinery & Aircraft Propulsion

Assignment – Propeller Design

By

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1 INTRODUCTION

Propeller design is crucial to an aircraft performance, efficiency, and safety. The propeller convert engine power into thrust, directly affecting take-off capability, climb rate, and fuel economy. A well-designed propeller minimizes aerodynamic losses, vibration, and noise while ensuring the tip speed remain below the transonic limit to avoid efficiency loss. Proper selection of blade geometry, aerofoil profile, and pitch distribution ensure an optimal balance between thrust generation, mechanical reliability, and overall efficiency, making propeller design a key factor in both performance and sustainability.



Figure 1: Propeller driven aircraft

This propeller is designed for the requirements given below is done by using MATLAB and the aerofoil is selected after analysing – aerofoils and chosen a high lift aerofoil. The justifications a discussed below.

The propeller,

- Be able to provide thrust for an aircraft of 1500kg
- Should be powered by 2-bladed twin propellers (connected to the engine, which rotates at 6000 rpm, through a 1:2 gear box) to take off at 220 km/hr speed at sea level within 1km distance,
- The aerodynamic drag of the aircraft at 220 km/h is 1.8kN
- The propeller is designed with blades of NACA blade sections.

- The speed of the blade tip is below the transonic speed and of 0.74 Mach with hub-to-tip width ratio of 3.
- 20 segments in discretized blades and no chamber.
- The size of the engine cover is 300 mm which determine the radius at the blade hub.

The design process involves determining the optimal propeller geometry, including blade radius, pitch distribution, and airfoil selection, while ensuring above requirements. The report also includes the estimation of thrust, torque, and power requirements, MATLAB-based computational routines, and a 3D propeller model developed in SOLIDWORKS to visualize the final design.

2 SPECIMEN HAND CALCULATION OF THE DESIGN

- Take off speed – 220 km/h
- Available runway length – 1 km
- After considering a factor of safety 1.5, 666.67 km of runway will be used for the take-off.
- Required acceleration – $a \text{ m/s}^2$

Assuming the acceleration is constant on the runway,

$$V^2 = U^2 + 2as$$

$$\left(\frac{220 \times 10^3}{60 \times 60}\right)^2 = 0^2 + 2 \times a \times 666.67$$

$$a = 2.8 \text{ m/s}^2$$

The incline angle at the take-off point is assumed as 8° .

Considering the forces on the aircraft in the direction of thrust,

$$F = ma$$

$$T = ma + W \sin 8 + D$$

$$T = 1500 \times 2.8 + 1500 \times .81 \sin 8 + 1800$$

$$T = 8047.93 \text{ N}$$

Approximately 8100 N of thrust is required for the aircraft to take off. Since the aircraft has twin propellers each propeller should be able to provide 4050 N.

- Engine RPM – 6000
- Gear box ratio – 1:2
- Propeller RPM – 3000
- Propeller blade tip Mach number – 0.74

$$V_{SOUND} = \sqrt{\gamma RT}$$

- Sound velocity at sea level – 331.2 m/s
- Propeller blade tip velocity - $0.74 \times 331.2 = 245.088 \text{ m/s}$
- Distance from centre to the blade tip – R

$$V = R\omega$$

$$245.1 = R \times 3000 \times \frac{2\pi}{60}$$

$$R = 0.78 \text{ m}$$

- Diameter of the propeller - $0.78 \times 2 = 1.56 \text{ m}$
- Hub diameter – 0.3 m
- Blade Length = $\frac{\text{Maximum diameter} - \text{hub diameter}}{2}$
Blade length = $(1.56 - 0.25)/2 = 0.63 \text{ m}$
- Number of specimens considered – 20
- Specimen length (dr) = $630 \text{ mm} / 20 = 31.50 \text{ mm}$

$$\text{Resultant velocity} = \sqrt{V_{\text{rotational speed at half blade span}}^2 + V_{\text{forward}}^2}$$

$$\text{Resultant velocity} = \sqrt{141.15^2 + 61.11^2}$$

$$\text{Resultant velocity} = 153.81 \text{ m/s}$$

$$\text{Reynold number} = \frac{\rho u c}{\mu} = \frac{1.2 \times 153.81 \times 0.115}{1.8 \times 10^{-5}} = 1\,179\,210$$

After selecting a suitable NACA aerofoil section for the propeller blade, below calculations were done. (Descriptive detail of aerofoil selection is provided in section 06)

- Selected aerofoil profile = NACA 6415
- Selected angle of attack = 13°
- $CL = 1.7910$
- $CD = 0.02515$

Element	dr	Radius m	Chord m	Va	Vt	Phi (rad)	u a	u theta	Pitch angle (Deg)
1	0.024	0.1658	0.2060	61.1100	52.0729	0.8651	0.1000	0.1000	62.5651
2	0.024	0.1973	0.1995	61.1100	61.9710	0.7784	0.1000	0.1000	57.5992
3	0.024	0.2288	0.1931	61.1100	71.8691	0.7047	0.1000	0.1000	53.3744
4	0.024	0.2603	0.1866	61.1100	81.7672	0.6418	0.1000	0.1000	49.7732
5	0.024	0.2918	0.1802	61.1100	91.6653	0.5880	0.1000	0.1000	46.6900
6	0.024	0.3233	0.1737	61.1100	101.5634	0.5417	0.1000	0.1000	44.0350
7	0.024	0.3548	0.1672	61.1100	111.4615	0.5015	0.1000	0.1000	41.7342
8	0.024	0.3863	0.1608	61.1100	121.3596	0.4665	0.1000	0.1000	39.7273
9	0.024	0.4178	0.1543	61.1100	131.2577	0.4357	0.1000	0.1000	37.9654
10	0.024	0.4493	0.1478	61.1100	141.1558	0.4086	0.1000	0.1000	36.4090
11	0.024	0.4808	0.1414	61.1100	151.0539	0.3844	0.1000	0.1000	35.0262
12	0.024	0.5123	0.1349	61.1100	160.9519	0.3629	0.1000	0.1000	33.7907
13	0.024	0.5438	0.1285	61.1100	170.8500	0.3435	0.1000	0.1000	32.6812
14	0.024	0.5753	0.1220	61.1100	180.7481	0.3260	0.1000	0.1000	31.6801
15	0.024	0.6068	0.1155	61.1100	190.6462	0.3102	0.1000	0.1000	30.7728
16	0.024	0.6384	0.1091	61.1100	200.5443	0.2958	0.1000	0.1000	29.9471
17	0.024	0.6699	0.1026	61.1100	210.4424	0.2826	0.1000	0.1000	29.1927
18	0.024	0.7014	0.0962	61.1100	220.3405	0.2705	0.1000	0.1000	28.5010
19	0.024	0.7329	0.0897	61.1100	230.2386	0.2594	0.1000	0.1000	27.8647
20	0.024	0.7644	0.0832	61.1100	240.1367	0.2492	0.1000	0.1000	27.2776

Table 1: Initial values for elements

Sample calculation for the first element

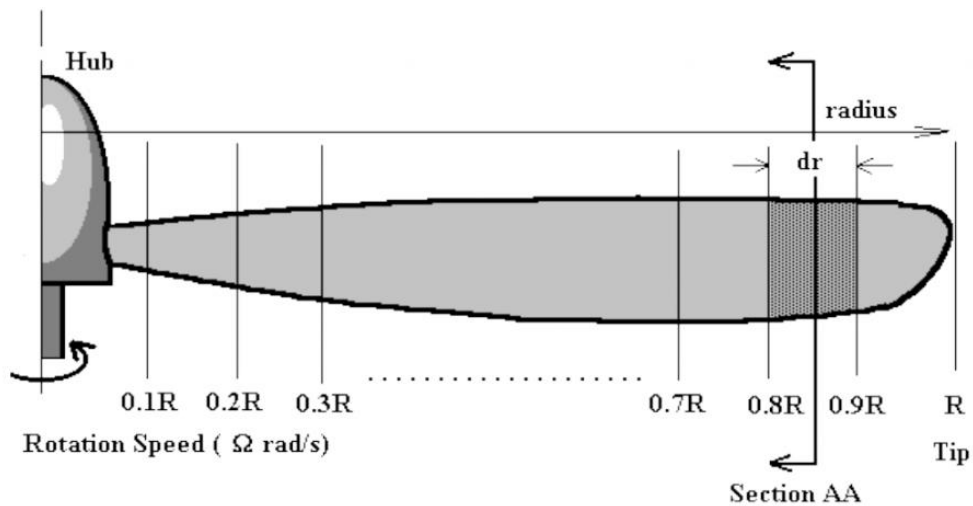


Figure 2: Blade section[1]

Mean radius for the blade element = 0.312

Initial guess for inflow factors-

$$u_a = 0.1, u_\theta = 0.1$$

New pitch angle calculation-

$$v_a = v_f + u_a = 62.11 \text{ m/s}$$

$$v_\theta = v_t - u_\theta = 51.973 \text{ m/s}$$

$$v_r = \sqrt{v_a^2 + v_\theta^2} = 80.99 \text{ m/s}$$

$$\phi = \tan^{-1}\left(\frac{v_a}{v_\theta}\right) = \tan^{-1}\left(\frac{62.11}{51.973}\right) = 0.8651 \text{ rad} = 49 \text{ deg}$$

$$\text{Pitch angle} = \theta = \phi + \alpha = 0.88 \text{ rad} = 62 \text{ deg}$$

From the blade element theory,

$$\Delta T = \frac{1}{2} \rho V_r^2 (C_L \cos \phi - C_D \sin \phi) B \cdot c \cdot dr = 112.82 \text{ N}$$

$$\Delta \tau = \frac{1}{2} \rho V_r^2 (C_D \cos \phi + C_L \sin \phi) B \cdot r \cdot c \cdot dr = 30.66 \text{ N}$$

These approximate values of thrust and torque can be used to get improved estimates of the new inflow factors (u_{a_new}) and (u_{θ_new}) using the following equations.

$$\Delta T = 4\pi\rho r(V_f + u_a)u_a dr$$

$$\Delta \tau = 4\pi\rho r^2(V_f + u_a)u_\theta dr$$

These steps are done iteratively until the u_a and u_θ values converge with 0.00001 tolerance.

After convergence,

$$v_a = 68.63 \text{ m/s}$$

$$v_\theta = 38.06 \text{ m/s}$$

3 EXCEL/MATLAB ROUTINE OF THE CALCULATION WITH USER INSTRUCTIONS

We have used MATLAB for the iterative calculation for the elements in propeller. Below provides the sample iteration results for the first element.

Iteration	ua	utheta	Va local	Vtheta local	dT b	dQ b	Phi (rad)
1	0.1000	0.1000	61.2111	51.9729	58.4464	11.7399	0.8668
2	10.1957	12.3555	71.3068	39.7174	44.9997	13.8451	1.0626
3	8.0890	15.0147	69.2001	37.0582	40.3381	12.9179	1.0791
4	7.3313	14.1643	68.4424	37.9086	41.1645	12.7377	1.0650
5	7.4667	13.9391	68.5778	38.1339	41.5338	12.8004	1.0633
6	7.5271	13.9954	68.6382	38.0775	41.4834	12.8156	1.0643
7	7.5188	14.0137	68.6300	38.0592	41.4546	12.8114	1.0645
8	7.5141	14.0101	68.6252	38.0629	41.4575	12.8101	1.0644
9	7.5146	14.0086	68.6257	38.0643	41.4597	12.8104	1.0644
10	7.5150	14.0088	68.6261	38.0641	41.4595	12.8105	1.0644
11	7.5149	14.0090	68.6261	38.0640	41.4594	12.8105	1.0644
12	7.5149	14.0089	68.6260	38.0640	41.4594	12.8105	1.0644

Table 2: Iterative process for first element

For the first element the convergence happens after 12 iterations with 0.00001 tolerance.

All the final values for the 20 elements are listed below.

Element	Iterations	Phi rad	Theta rad	dT	d tau
1	12	1.0644	1.2389	41.4594	12.8105
2	11	0.9530	1.1276	57.2470	16.3753
3	10	0.8615	1.0361	73.9068	20.2693
4	10	0.7851	0.9596	91.4627	24.4689
5	10	0.7204	0.8949	109.8291	28.9347
6	9	0.6649	0.8395	128.8562	33.6163
7	9	0.6169	0.7915	148.3543	38.4554
8	9	0.5750	0.7495	168.1089	43.3881
9	8	0.5380	0.7125	187.8892	48.3463
10	8	0.5051	0.6796	207.4533	53.2586
11	8	0.4756	0.6501	226.5514	58.0515
12	8	0.4490	0.6236	244.9286	62.6501
13	8	0.4249	0.5994	262.3259	66.9784

14	8	0.4029	0.5774	278.4815	70.9604
15	7	0.3827	0.5572	293.1314	74.5202
16	7	0.3640	0.5386	306.0098	77.5830
17	7	0.3467	0.5213	316.8496	80.0751
18	7	0.3306	0.5051	325.3830	81.9250
19	7	0.3155	0.4901	331.3408	83.0635
20	7	0.3014	0.4759	334.4537	83.4248

Table 3: Final approximation values for 20 elements

For a single propeller,

Total thrust = 4134.022 N

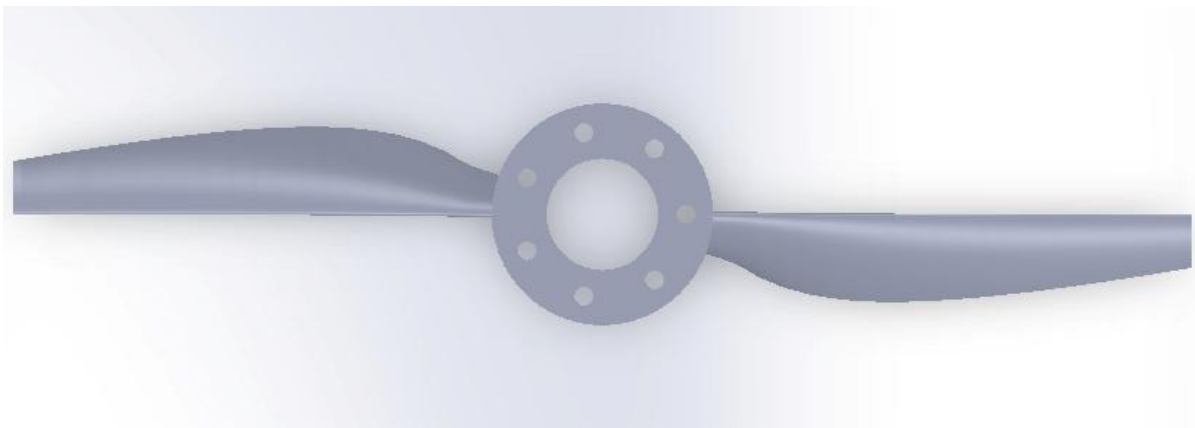
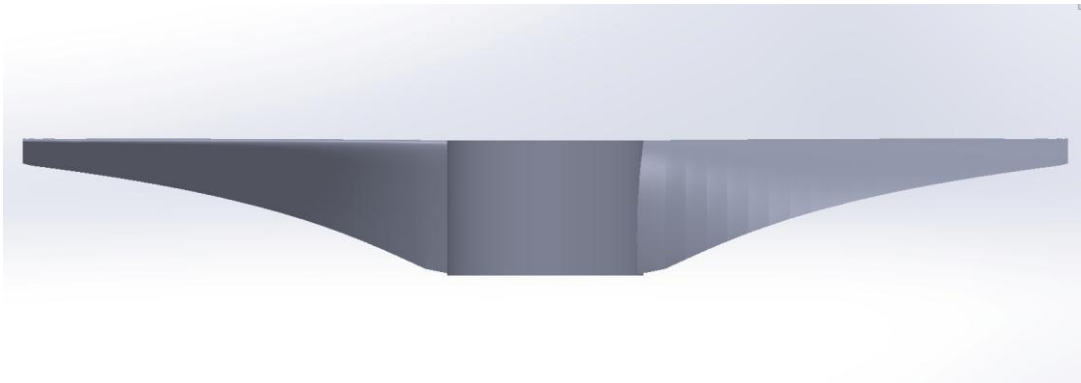
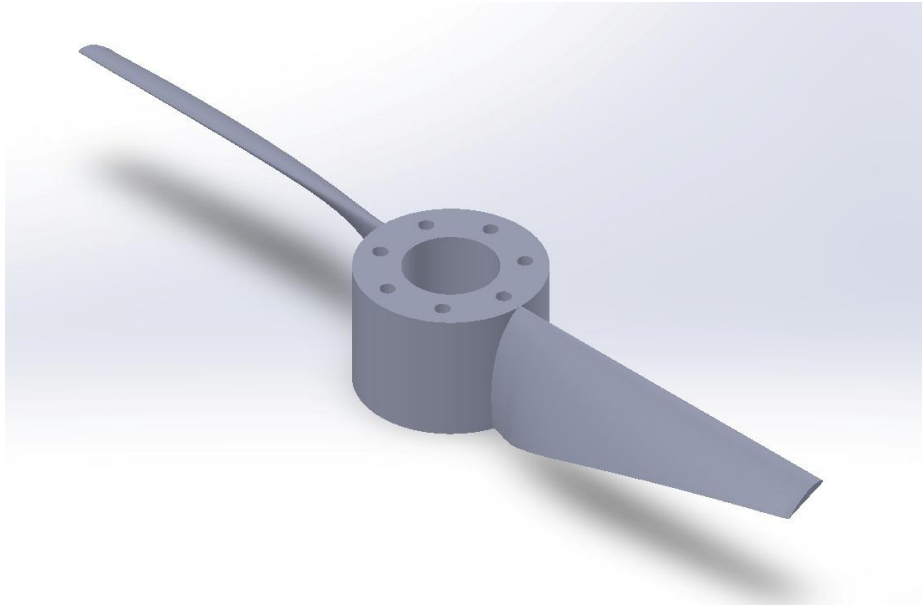
Total Torque = 1059.1552 Nm

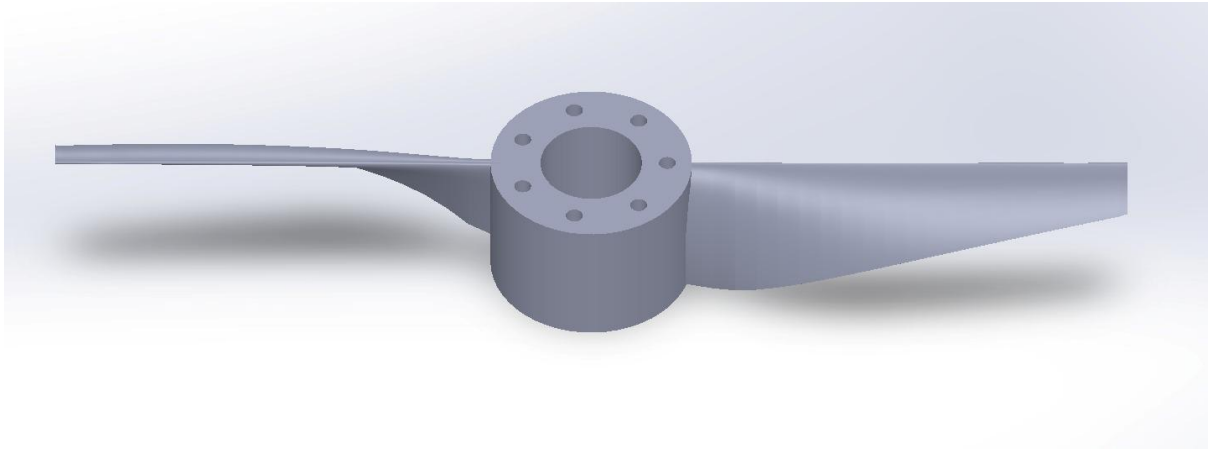
For both propellers,

Total thrust = 8264.044 N

Total Torque = 2118.3104 Nm

4 3D DESIGN OF THE PROPELLER ON SOLIDWORKS





5 THE CALCULATION OF ENGINE POWER REQUIRED

$$T = \Sigma \Delta T = 8264.044 \text{ N}$$

$$\tau = \Sigma \Delta \tau = 2118.31 \text{ Nm}$$

Advanced ratio

$$J = \frac{V}{ND} = 0.7833$$

Thrust coefficient

$$C_T = \frac{T}{\rho N^2 D^4} = 0.227774$$

Torque Coefficient

$$C_\tau = \frac{\tau}{\rho N^2 D^5} = 0.037402$$

Power Coefficient

$$C_p = \frac{\tau}{\rho N^3 D^5} = 0.235003$$

Efficiency of the propeller

$$\eta_{prop} = \frac{J C_T}{2\pi C_\tau} = 0.759248$$

Required Engine Power,

$$P = \frac{TV}{\eta_{prop}} = 332743.41 \text{ W} = 332.743 \text{ kW}$$

6 THE PROFILE NUMBER OF THE AERO FOIL YOU SELECTED FROM NACA WITH THE JUSTIFICATIONS

Below listed the commonly used NACA profiles for propellers and their maximum lift coefficient values with respective angle of attack and drag coefficient. We have chosen the maximum lift angle of attack values as the propeller design is mainly focused on creating maximum thrust which improves the fuel efficiency and efficient take-off.[2] [3]

Aerofoil	Max CL AoA	Max CL	CD at max CL AoA	Comments
NACA 4412	15.5	1.5171	0.06031	Commonly used in general aviation propellers. Provide high lift in moderate Reynolds numbers.
NACA 4415	15.25	1.4697	0.04399	Creates slightly high lift and better structural stiffness for blades. Efficient in takeoff and climb.
NACA 6415	13	1.7910	0.02515	Give very high lift with safe stall condition. Used commonly in general aviation propellers.
NACA 23012	14.25	1.4299	0.03162	Widely used in aircraft. Efficient thrust with smooth stall.
NACA23015	15.5	1.4303	0.04107	Balanced between strength and aerodynamic efficiency. Often used in general purpose propellers.
NACA 16-012	10.0	1.0099	0.0401	Suitable for high efficiency and low-drag applications.
NACA 64-415	16.50	1.3482	0.07664	Used in modern propellers for lower noise in cruise condition.

Table 4: Selected aerofoils' details

From above aerofoils, we have chosen **NACA 6415** aerofoil section, as it gives high aerodynamic efficiency and high lift coefficient in smaller attack angle. This is favourable in avoiding stall conditions of the aerofoil. And also, it generates low drag value at the relevant attack angle.

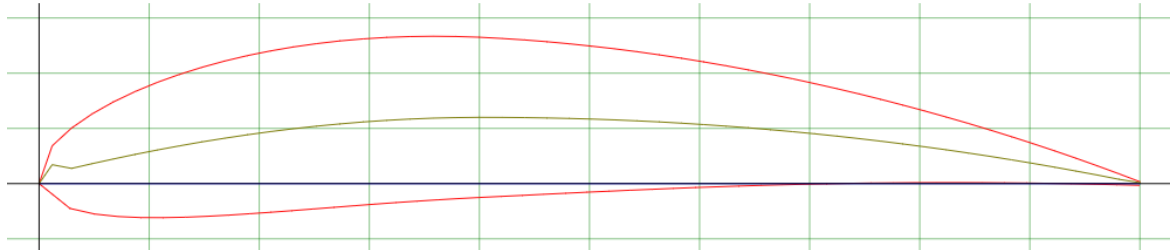


Figure 3: NACA 6415 geometry

alpha	CL	CD
-5	0.1417	0.00725
-4	0.2554	0.00711
-3	0.3691	0.00704
-2	0.4826	0.00702
-1	0.5963	0.00701
0	0.7094	0.00706
1	0.8225	0.00711
2	0.9353	0.00716
3	1.0468	0.00682
4	1.153	0.00653
5	1.2614	0.00694
6	1.3674	0.0074
7	1.4672	0.00802
8	1.552	0.00902
9	1.6293	0.01039
10	1.6931	0.0124
11	1.736	0.01559
12	1.7661	0.01988
13	1.791	0.02515
14	1.8152	0.03131
15	1.8334	0.03878
16	1.8417	0.04793
17	1.8432	0.05845
18	1.8348	0.07058
19	1.8198	0.08387
20	1.8027	0.09776

Table 5: NACA 6415 values

7 REFERENCES

- [1] “Blade Element Propeller Theory | Aerodynamics for Students.” Accessed: Oct. 09, 2025. [Online]. Available: <https://www.aerodynamics4students.com/propulsion/blade-element-propeller-theory/>
- [2] W. M. Wan Mohamed, N. P. Ravindran, and P. Rajendran, “A CFD Simulation on the Performance of Slotted Propeller Design for Various Airfoil Configurations,” *CFD Lett.*, vol. 13, no. 3, pp. 43–57, Mar. 2021, doi: 10.37934/cfdl.13.3.4357.
- [3] E. P. Hartman and D. Biermann, “The Aerodynamic Characteristics of Full-Scale Propellers Having 2, 3, and 4 Blades of Clark Y and R.A.F. 6 Airfoil Sections.” Jan. 01, 1938. Accessed: Oct. 09, 2025. [Online]. Available: <https://ntrs.nasa.gov/citations/19930091715>

APPENDICES

CAD File - <https://dms.uom.lk/s/QGGAKC7k3Z9tQb6>

MATLAB Code - <https://dms.uom.lk/f/57319301>